Sardar Patel Institute of Technology
Department of Computer Engineering · T.E. Mini Project · AY 2026–27

GROUP 86

Dynamic Timetable

A role-based timetable system that reads the college's PDF and answers questions in plain English

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GUIDE
Prof. Anant Nimkar

AGENDA

What we will cover

01

The problem

Why a timetable PDF is hard to use

02

Our solution

Results, architecture and stack

03

How it works

Extraction, clashes, query planner

04

The product

Screens for students, staff, admin

05

Results

Testing, performance, comparison

06

What’s next

Limitations and future scope

THE PROBLEM

Everyone reads a 10-page PDF by hand

“Where is my next class?”

Find the page, row, column and your lab batch

“Is room 508 free at 2?”

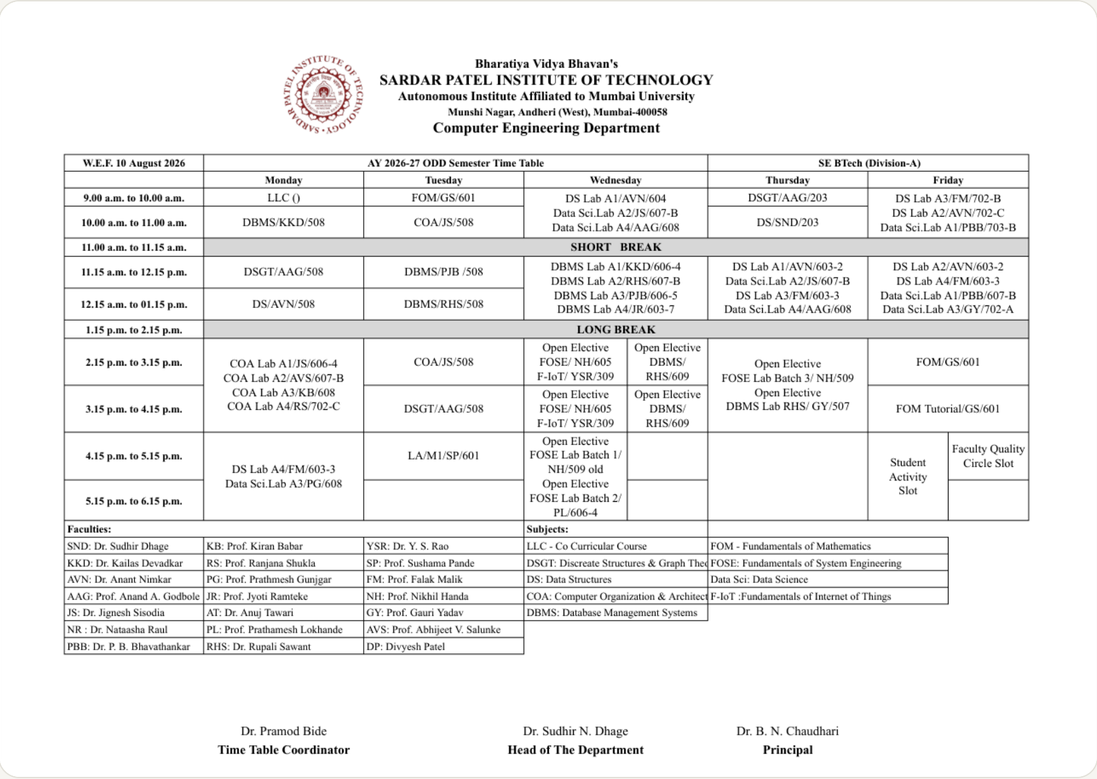
Check every page for every room

Clashes found too late

A teacher in two rooms is noticed in the corridor

Which version is official?

Old and revised PDFs circulate together



Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
Autonomous Institute Affiliated to Mumbai University
Munshi Nagar, Andheri (West), Mumbai-400058
Computer Engineering Department

W.E.F. 10 August 2026	AY 2026-27 ODD Semester Time Table				SE BTech (Division-A)	
	Monday	Tuesday	Wednesday		Thursday	Friday
9.00 a.m. to 10.00 a.m.	LLC ()	FOM/GS/601	DS Lab A1/AVN/604 Data Sci.Lab A2/JS/607-B Data Sci.Lab A4/AAG/608		DSGT/AAG/203	DS Lab A3/FM/702-B DS Lab A2/AVN/702-C Data Sci.Lab A1/PBB/703-B
10.00 a.m. to 11.00 a.m.	DBMS/KKD/508	COA/JS/508			DS/SND/203	
11.00 a.m. to 11.15 a.m.	SHORT BREAK					
11.15 a.m. to 12.15 p.m.	DSGT/AAG/508	DBMS/PJB /508	DBMS Lab A1/KKD/606-4 DBMS Lab A2/RHS/607-B DBMS Lab A3/PJB/606-5 DBMS Lab A4/JR/603-7		DS Lab A1/AVN/603-2 Data Sci.Lab A2/JS/607-B DS Lab A3/FM/603-3 Data Sci.Lab A4/AAG/608	DS Lab A2/AVN/603-2 DS Lab A4/FM/603-3 Data Sci.Lab A1/PBB/607-B Data Sci.Lab A3/GY/702-A
12.15 a.m. to 01.15 p.m.	DS/AVN/508	DBMS/RHS/508				
1.15 p.m. to 2.15 p.m.	LONG BREAK					
2.15 p.m. to 3.15 p.m.	COA Lab A1/JS/606-4 COA Lab A2/AVS/607-B COA Lab A3/KB/608 COA Lab A4/RS/702-C	COA/JS/508	Open Elective FOSE/ NH/605 F-IoT/ YSR/309	Open Elective DBMS/ RHS/609	Open Elective FOSE Lab Batch 3/ NH/509 Open Elective DBMS Lab RHS/ GY/507	FOM/GS/601
3.15 p.m. to 4.15 p.m.		DSGT/AAG/508	Open Elective FOSE/ NH/605 F-IoT/ YSR/309	Open Elective DBMS/ RHS/609		FOM Tutorial/GS/601
4.15 p.m. to 5.15 p.m.	DS Lab A4/FM/603-3 Data Sci.Lab A3/PG/608	LA/M1/SP/601	Open Elective FOSE Lab Batch 1/ NH/509 old			Student Activity Slot / Faculty Quality Circle Slot
5.15 p.m. to 6.15 p.m.			Open Elective FOSE Lab Batch 2/ PL/606-4			

Faculties:			Subjects:	
SND: Dr. Sudhir Dhage	KB: Prof. Kiran Babar	YSR: Dr. Y. S. Rao	LLC - Co Curricular Course	FOM - Fundamentals of Mathematics
KKD: Dr. Kailas Devadkar	RS: Prof. Ranjana Shukla	SP: Prof. Sushama Pande	DSGT: Discreate Structures & Graph Theo	FOSE: Fundamentals of System Engineering
AVN: Dr. Anant Nimkar	PG: Prof. Prathmesh Gunjgar	FM: Prof. Falak Malik	DS: Data Structures	Data Sci: Data Science
AAG: Prof. Anand A. Godbole	JR: Prof. Jyoti Ramteke	NH: Prof. Nikhil Handa	COA: Computer Organization & Architect	F-IoT :Fundamentals of Internet of Things
JS: Dr. Jignesh Sisodia	AT: Dr. Anuj Tawari	GY: Prof. Gauri Yadav	DBMS: Database Management Systems	
NR : Dr. Nataasha Raul	PL: Prof. Prathamesh Lokhande	AVS: Prof. Abhijeet V. Salunke		
PBB: Dr. P. B. Bhavathankar	RHS: Dr. Rupali Sawant	DP: Divyesh Patel		

Dr. Pramod Bide
Time Table Coordinator

Dr. Sudhir N. Dhage
Head of The Department

Dr. B. N. Chaudhari
Principal

Page 1 of the real CE timetable (SE Division A)

WHY IT IS HARD FOR SOFTWARE TOO

The real PDF breaks simple parsers

Stacked lab cells

Four batches (A1–A4) with their own teacher and lab inside one cell

Page-wise legends

KKD, DBMS, LLC are expanded only at the bottom of each page

Wrong time labels

The noon row is printed “12.15 a.m. to 01.15 p.m.”

Combined pages

One BE page of electives serves divisions A to D

Missing details

LLC slots have no teacher or room; one M.Tech row has no time

Spelling varies

“Sushama” and “Sushma” Pande; “Discreate” Structures

OBJECTIVES

Nine goals we set — and met

01	At least 95% of classes extracted and verified
02	Accept PDF, scanned image, Excel and Word
03	Five roles: Student, Professor, HOD, Principal, Admin
04	Plain-English questions, not a fixed list
05	Detect teacher, room and batch clashes
06	Keep official and personal timetables separate
07	Answer any question in under 300 ms
08	Android, iOS and web, with a free live demo
09	Every automated test passing

All nine are met — the results section shows the evidence for each.

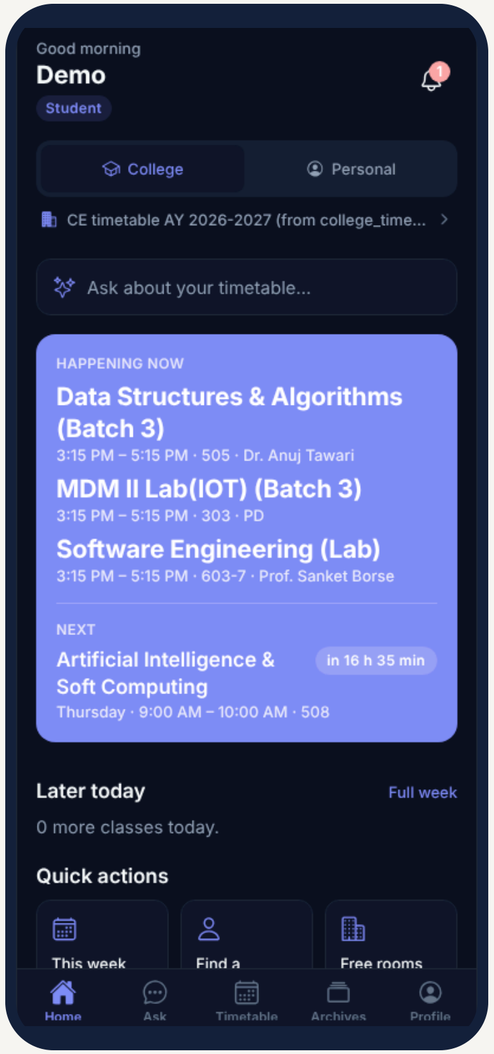
OUR SOLUTION

Upload once. Ask anything.

1 **Admin uploads the PDF**
The system reads every cell, legend and lab batch

2 **It checks and publishes**
Clashes are flagged before anyone sees the timetable

3 **Everyone asks in English**
“When is my last class on Friday?” — answered in 40 ms



HEADLINE RESULTS

On the college’s real timetable

99.8%

classes verified

452 of 453 extracted classes

2

genuine clashes

out of 542 raw time overlaps

~40 ms

median answer time

target was under 300 ms

1.65 s

to read 10 pages

whole PDF, upload to verified

199

automated tests

179 back end + 20 app, all passing

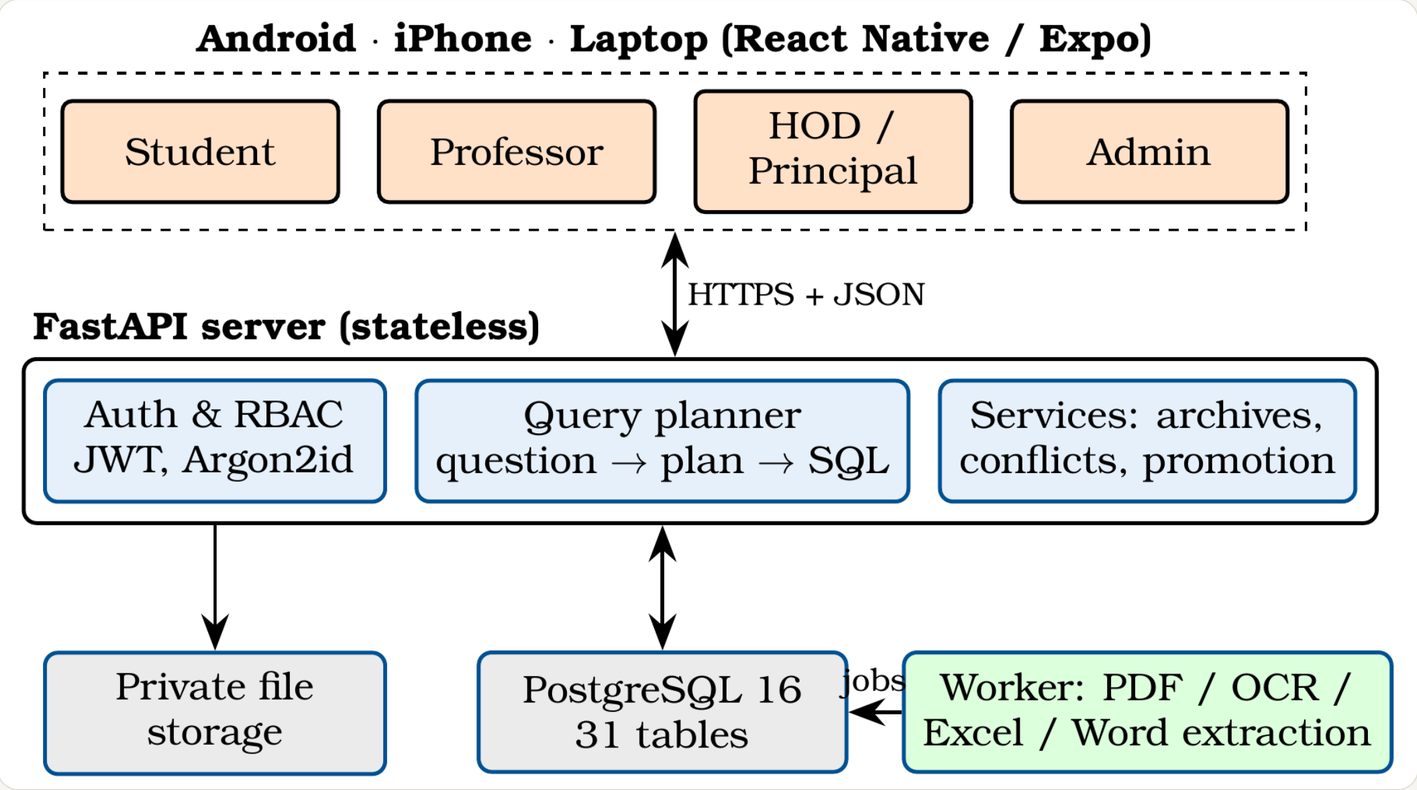
Rs. 0

running cost

open-source stack, free cloud tier

ARCHITECTURE

One API, one database, one app



Client
One Expo codebase for Android, iOS and web

API
FastAPI with JWT, roles and rate limits

Worker
Extracts uploads from a Postgres job queue

Database
PostgreSQL 16, 31 tables, the single source of truth

TECHNOLOGY STACK

Open-source, chosen for a reason

BACK END

FastAPI · Python 3.13

Typed, async, auto-generated API docs

DATABASE

PostgreSQL 16

SQLAlchemy 2 + Alembic migrations; also the job queue

APP

React Native · Expo 58

TypeScript strict, Expo Router, TanStack Query

EXTRACTION

PyMuPDF · Tesseract

Plus openpyxl and python-docx for Excel and Word

SECURITY

Argon2id · JWT

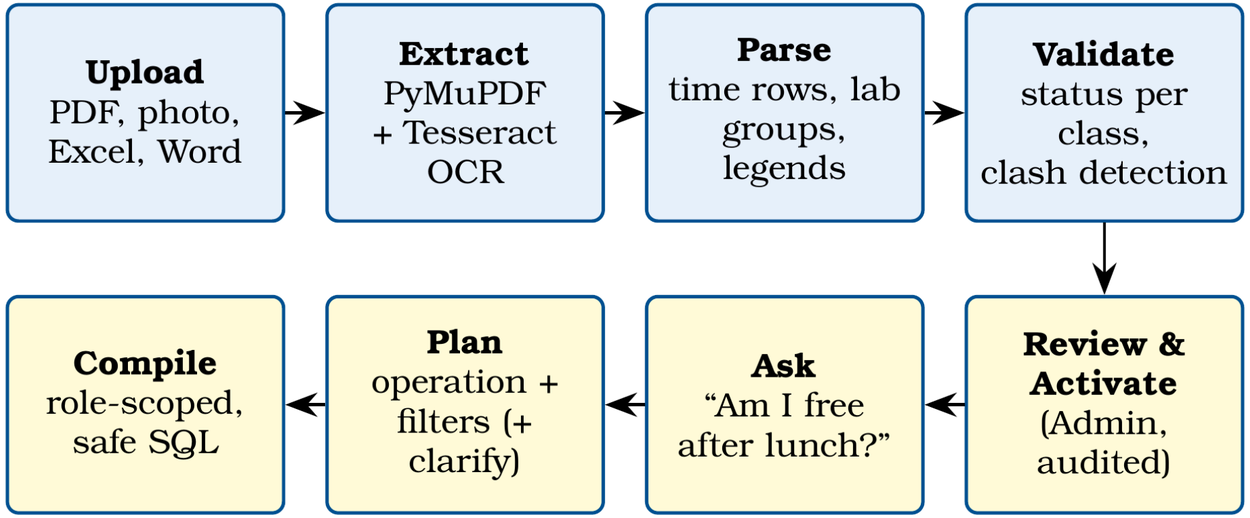
Rotating refresh tokens, role-based access

DEPLOYMENT

Docker · Render

One blueprint file, free tier, HTTPS included

From PDF to verified classes



Read the grid

Words with positions, ruled lines become cells

Expand codes

Each page’s legend maps KKD → teacher, DBMS → subject

Split lab cells

A1–A4 each become their own class row

Score confidence

Unsure rows are flagged for admin review

Fixing wrong time labels by backtracking

TIME COLUMN OF THE PDF

“11.15 to 12.15”	**11:15–12:15**
“12.15 a.m. to 01.15 p.m.”	**12:15–13:15**
“1.15 to 2.15”	**13:15–14:15**
“2.15 to 4.15”	**14:15–16:15**

The noon row is proven by its neighbours, not assumed

1 · Candidates

Each label gives a few readings (a.m. / p.m.)

2 · Search

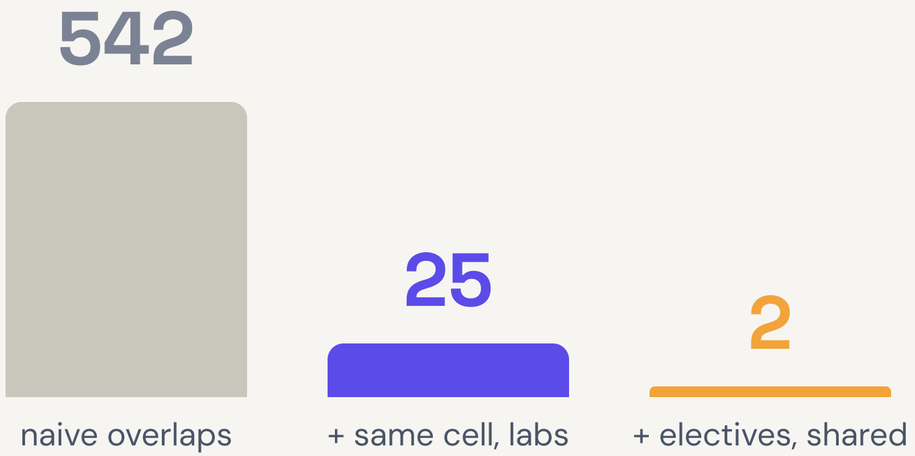
Keep only assignments where each row is 7:00–21:00 and starts after the last one ends

3 · Certainty

A row is certain only if it is the same in every valid assignment — otherwise it goes to review

Bounded at 64 assignments → O(64·r) for r rows

Finding real clashes, not false alarms



Both remaining clashes were confirmed by hand against the PDF.

Group → sort → sweep

Group by teacher, room or batch per day; sort by start; stop as soon as the next class starts after this one ends

Overlap rule

$a.start < b.end \land b.start < a.end$

End-exclusive: 10–11 and 11–12 do not clash

O(n log n + m)

instead of $O(n^2)$ pairwise checks

HOW IT WORKS · 4 OF 4 · QUERY PLANNER

A question becomes a typed plan

1 · QUESTION

“Which day do I finish earliest?”

Words are matched against the timetable’s own vocabulary — teachers, rooms, subjects, batches

→

2 · PLAN

```
operation: PER_DAY
filters: me
metric: LAST_END
sort: EARLIEST
```

→

3 · ANSWER

The day, and when its last class ends

Compiled to parameterised SQL, run on the official data

17 operations · count, list, next, first/last, free slots, per day, who teaches…

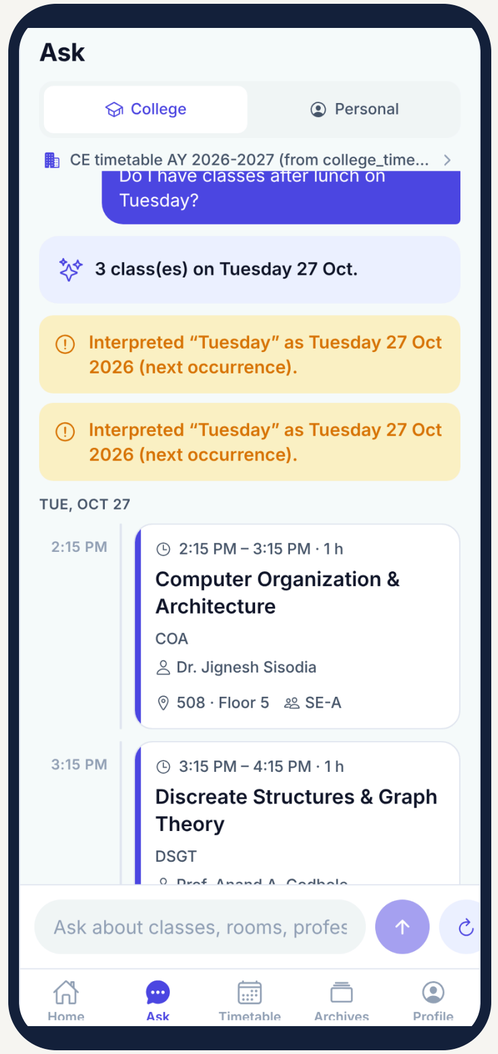
6 time windows · at, after, before, between, after lunch, now

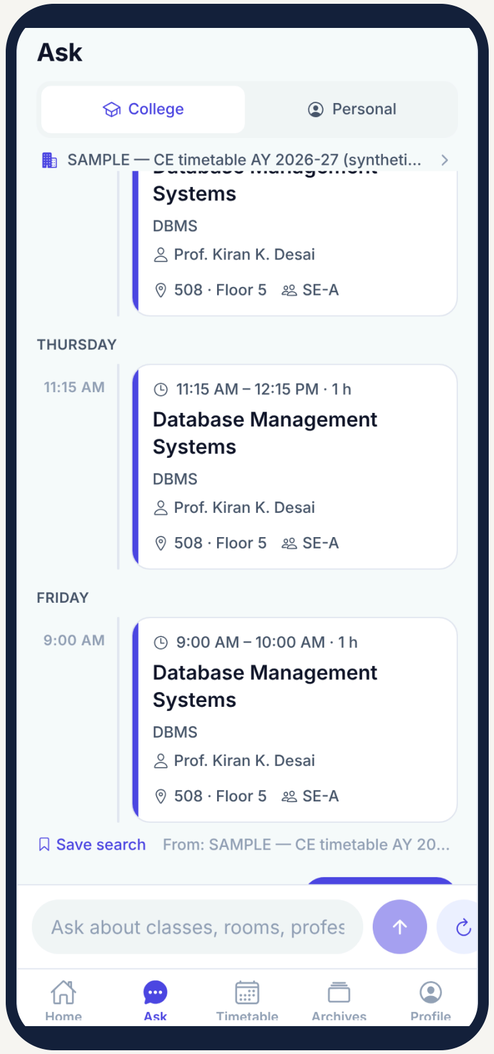
NOT A FIXED LIST OF QUESTIONS

Pieces combine freely

- “How many labs do I have this week?”
- “What do I have after lunch on Tuesday?”
- “Is room 508 free at 2 pm?”
- “Who teaches DBMS?”
- “Where is Nimkar sir on Monday?”
- “Which is my busiest day?”
- “Free slots on Thursday”
- “What comes after OS lab?”

Every answer card shows the plan it ran, so the result can be checked.





WHY NOT JUST USE AN LLM?

Exact answers, never guesses

Deterministic

Same question, same answer, straight from the database

Injection-safe

Closed vocabulary and parameterised SQL — no free-form query text

Asks when unsure

“At 2” → AM or PM? Two teachers match → which one?

Free and offline

No API key, no per-question cost, no student data sent out

An LLM could later *suggest* plans — still checked by the same validator.

UNDERSTANDING NAMES

Ask the way people actually talk

TEACHERS

AVN · Anant · Nimkar sir

Code, first name, surname or “X sir / ma’am” all reach the same person

SPELLING VARIANTS

Sushama = Sushma Pande

Entries sharing one faculty code are merged into one person

SUBJECTS

discrete · “Discreate”

Partial words and one-letter typos match (edit distance ≤ 1)

BATCHES

A2 → also SE Div A

A lab batch includes its parent division’s lectures

Exact match first, then unique word, then fuzzy **suggestions** — it never silently picks the wrong teacher.

SECURITY AND ROLES

Open to view, locked to change

Action	Who
View any official timetable	Everyone
Ask questions	Everyone
Personal timetable	Owner only
See clashes	HOD, Principal, Admin
Upload, correct, publish	Admin only

Argon2id passwords
Memory-hard hashing, login rate limits

Short-lived tokens
15-min access token; 30-day refresh stored only as a hash

Theft detection
Reusing an old refresh token revokes the whole family

Audit trail
Every admin change is logged, append-only

DATABASE DESIGN

The rules live in the schema

31
tables, 3NF

3
Alembic migrations

570
rows for the real PDF

12 MB
whole database

CHECK constraints
day 1–7, end > start, confidence 0–1, known status values

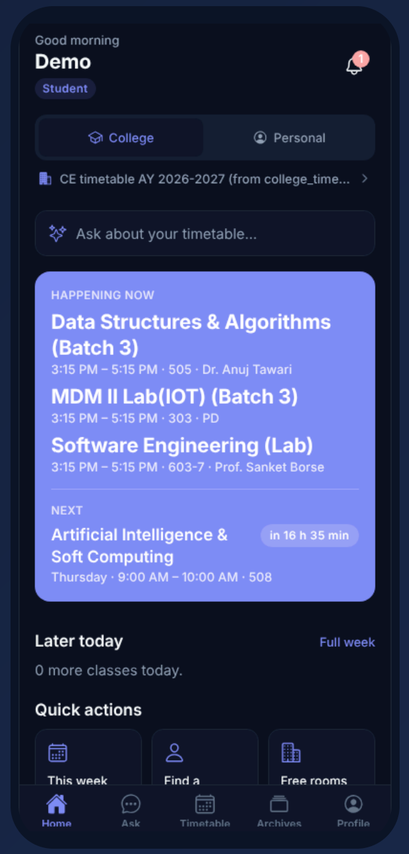
Partial unique indexes
one current placement per student, idempotent uploads

Composite foreign keys
a user can only point to their *own* personal timetable

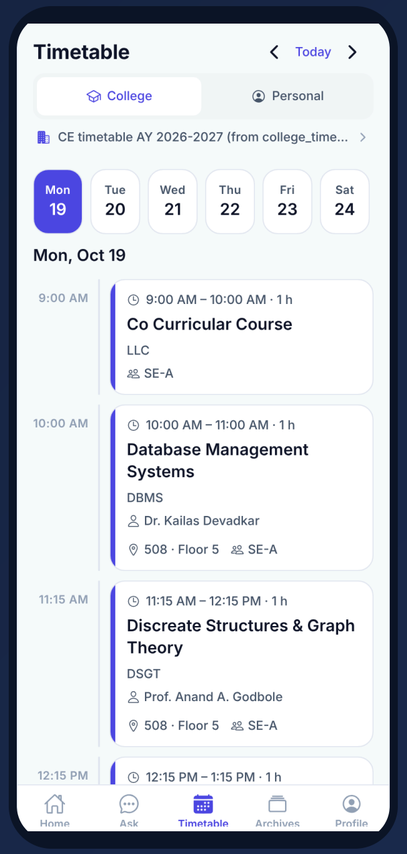
Provenance on every row
raw text, page and confidence kept for review

THE PRODUCT · STUDENT

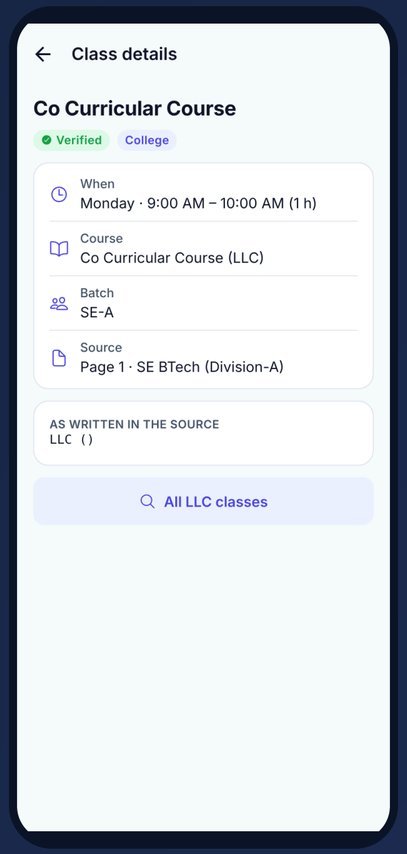
My day at a glance



Home · next class



Monday timetable



Class details

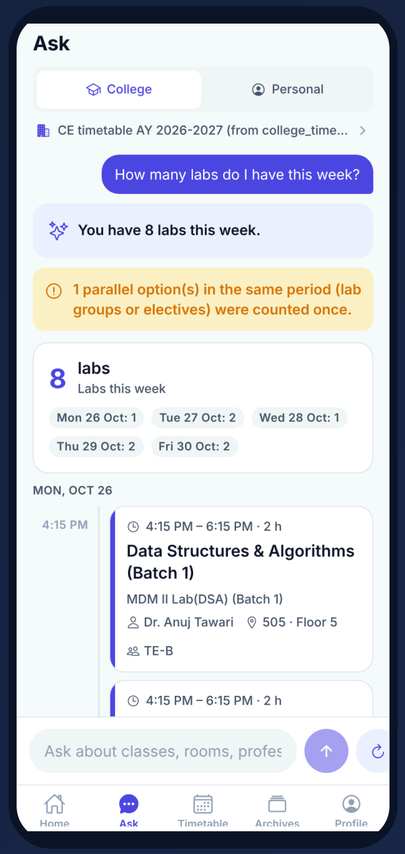
Only **their** batch's classes, lab batch included

Full teacher name, room and time

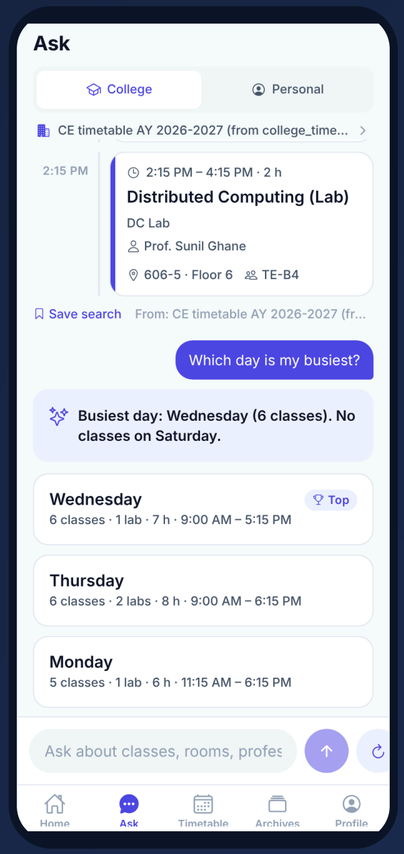
Source page of the PDF for trust

Works offline with the last copy

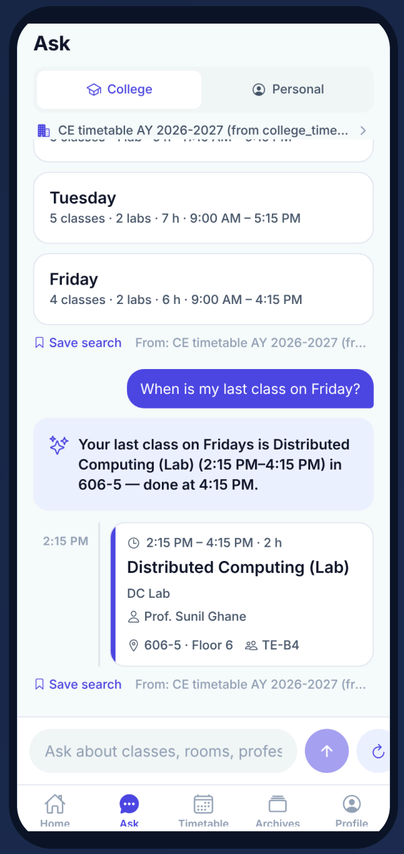
Count, rank, find — in one box



"How many…?" · COUNT



"Busiest day?" · PER_DAY



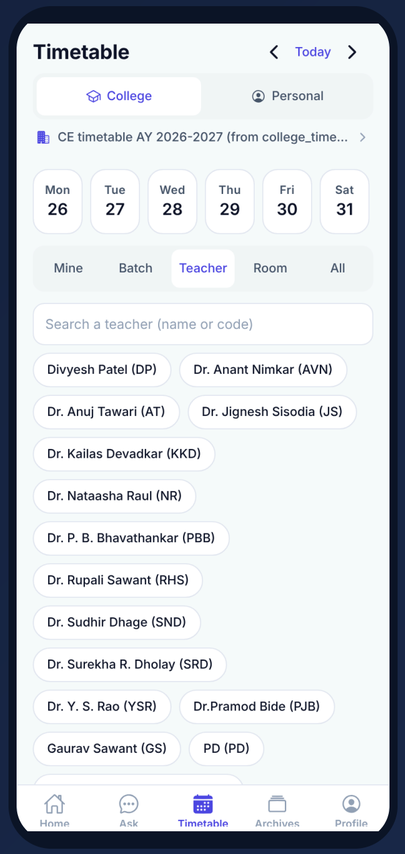
"Last class?" · LAST

Answer cards match the question: a number, a ranked list or a class

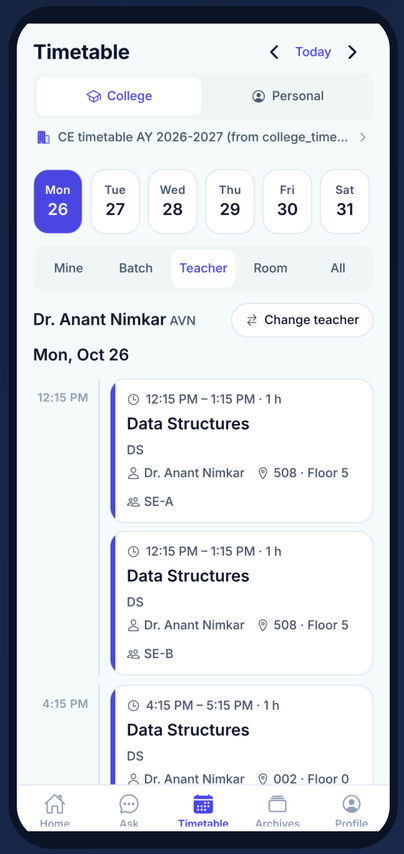
Suggested questions to start from

Tap "?" to see the **plan** that was run

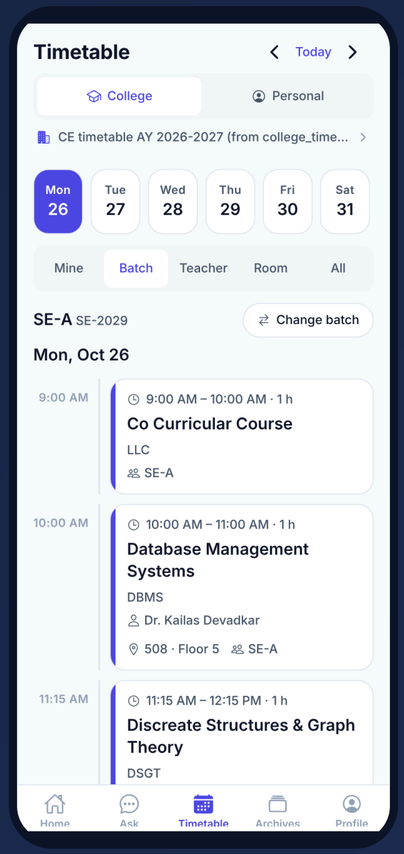
Any teacher, batch or room in two taps



Pick a teacher



Their Monday



Another batch

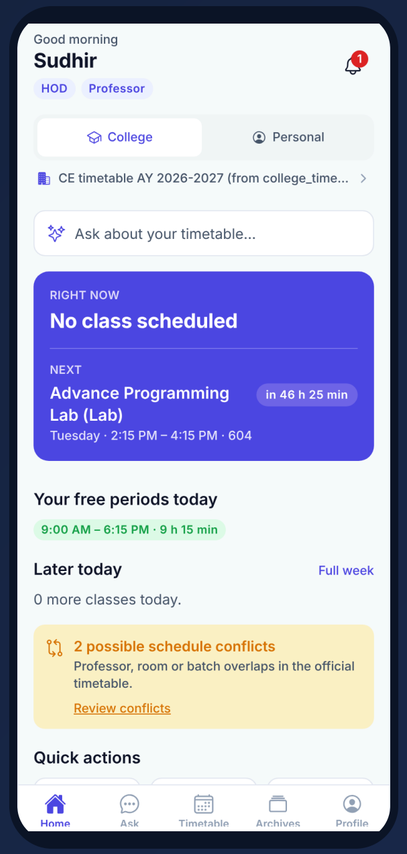
Mine · Batch · Teacher · Room · All

A student can find a teacher for doubts

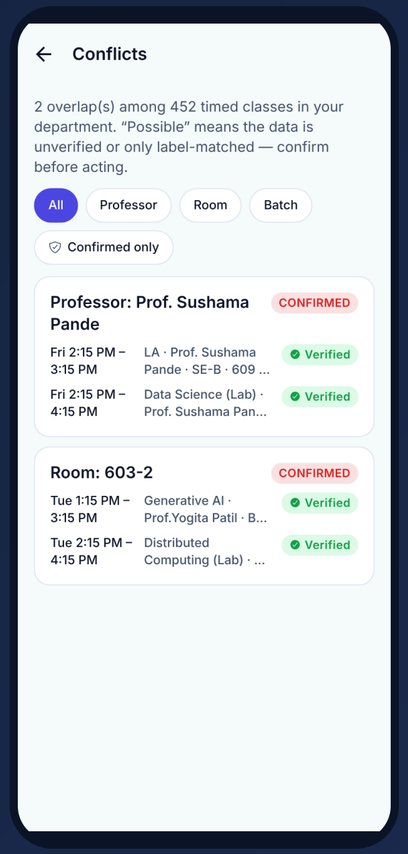
A teacher can check a class before swapping

Searchable list, filtered by department

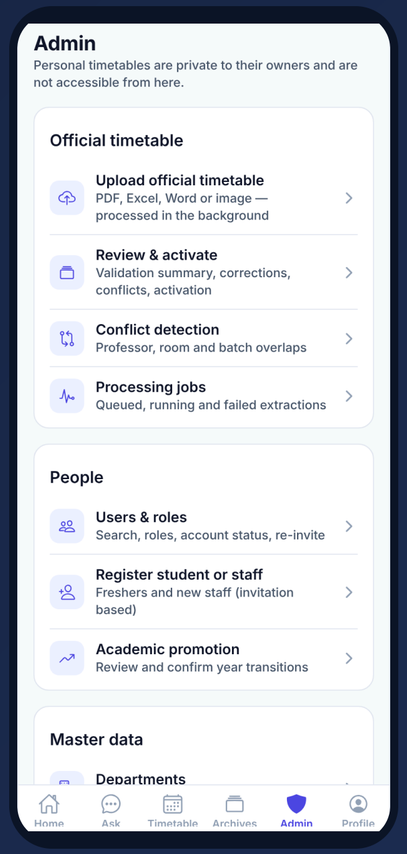
Upload, review, fix clashes, publish



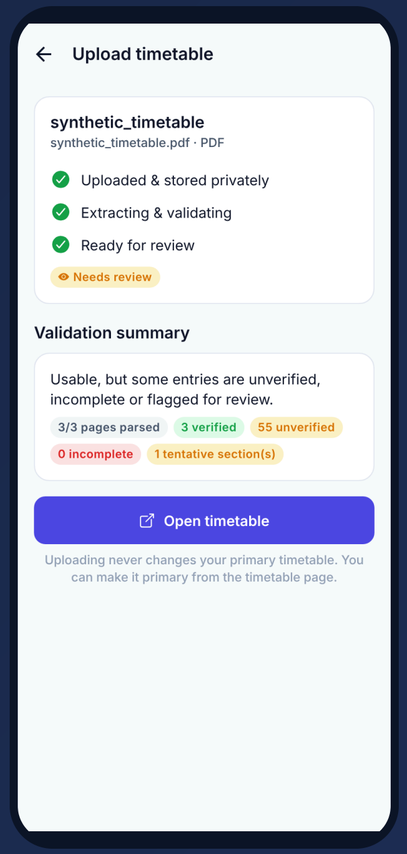
HOD home



Clashes



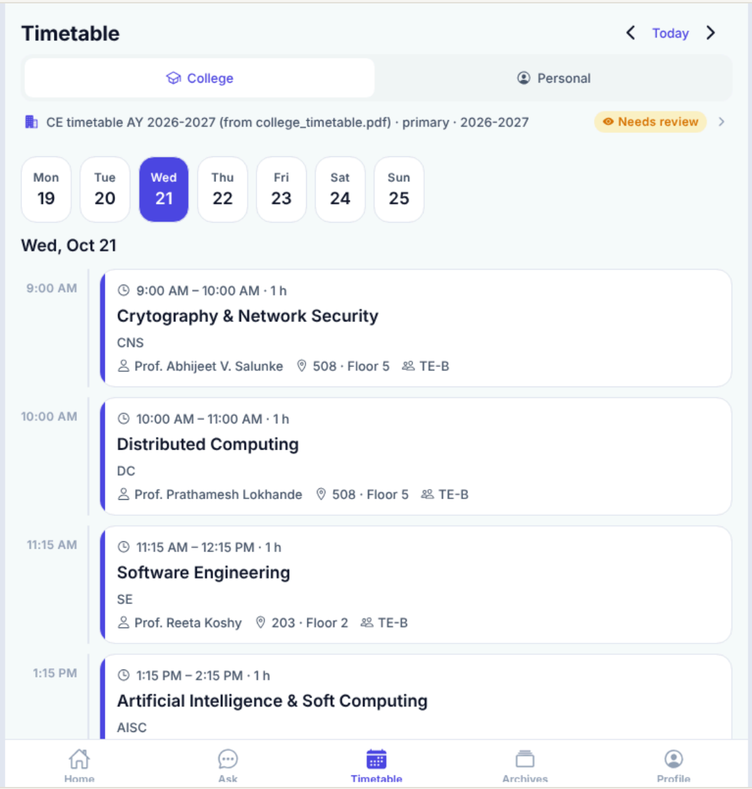
Admin hub



Upload processed

THE PRODUCT · WEB AND CLOUD

Same app on a laptop, live on the cloud



One codebase

Expo builds Android, iOS and the web from the same TypeScript

One-file deployment

render.yaml + Dockerfile: API, worker and database on the free tier

Runs locally too

One script starts Postgres, API and app for a demo

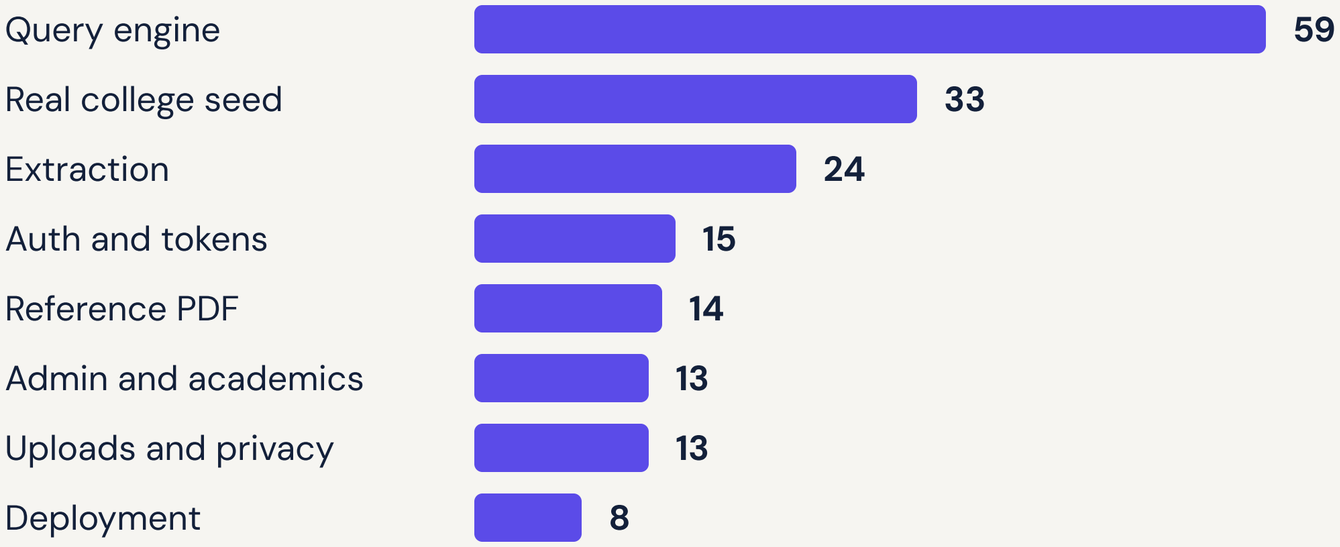
Small footprint

API worker ~115 MB RAM, database 12 MB

TESTING

244 checks, all green

179 BACK-END TESTS (PYTEST, REAL POSTGRESQL)



20
app tests (Jest) + TypeScript strict

45
end-to-end checks against the running API (24 college + 21 smoke)

Includes privacy attacks: another user's personal timetable returns **404**

PERFORMANCE · LOAD TEST

Fast alone, steady under load

Clients	req/s	p50 ms	p95 ms
1	18.6	49	57
2	25.4	75	90
4	46.5	64	90
8	**60.4**	**90**	**141**
16	54.1	183	354
32	57.4	410	635

37–47 ms

median per question type, one user (p95 ≤ 54 ms)

0

wrong answers even at 32 parallel clients

Throughput peaks at 8 clients; beyond that requests queue rather than fail. Measured on development hardware.

COMPARISON

What existing tools leave out

Criterion	Manual PDF	FET / aSc	Tabula	LLM SQL	Ours
Reads the published PDF	—	No	Partly	No	**Yes**
Understands labs and legends	Human	Yes	No	Depends	**Yes**
Plain-English questions	No	No	No	Yes	**Yes**
Exact, injection-safe answers	—	—	—	No	**Yes**
Clash check after publishing	No	At creation	No	No	**Yes**
Role-based mobile app	No	Limited	No	Varies	**Yes**
Cost	Free	Free/Paid	Free	Paid API	**Free**

FET / aSc *create* timetables; we work on the timetable after it is published — the two are complementary.

SYLLABUS APPLIED

Where our coursework shows up

DBMS

3NF schema, constraints, indexes, transactions and row locks

OS

Producer–consumer job queue, mutual exclusion with SKIP LOCKED

Networks

REST over HTTPS, JSON, single-flight token refresh

Data Structures

Hash maps, sets, grids and queues in extraction and clashes

Algorithms

Backtracking for time rows, sort-and-sweep in O(n log n)

Software Engg.

SRS, traceability, 244 automated checks, Git history

Discrete Maths

Interval overlap rule, relations, role matrix

Cloud

Docker, PaaS, infrastructure as code with render.yaml

10.5k lines of Python

5.5k lines of TypeScript

HONEST LIMITS · WHAT'S NEXT

What we would do with more time

CURRENT LIMITATIONS

- Accuracy not yet measured against hand-labelled pages
- Load-tested only on development hardware
- No Play Store / TestFlight build or push notifications yet
- Some short forms (e.g. "AI") not recognised; bare "at 2" asks AM/PM

FUTURE SCOPE

- College single sign-on (Google Workspace)
- "What changed?" alerts sent only to affected users
- Optional LLM that *suggests* plans, still validated
- CSV import, room-use analytics, exam timetables

THANK YOU

Questions?

Happy to run a live demo — ask the app anything about the CE timetable.

Shivam Mulani · Gurshaan Nandrajog · Harshvarddhan Waghela

Guide: Prof. Anant Nimkar · HOD: Dr. Sudhir Dhage · Sardar Patel Institute of Technology

Source code on GitHub